Aliah University
Department of Electrical Engineering
B. Tech. III semester Examination December -2022

Sub: Circuit Theory & Networks

Full Marks: 80 — Code- EENUGOE01

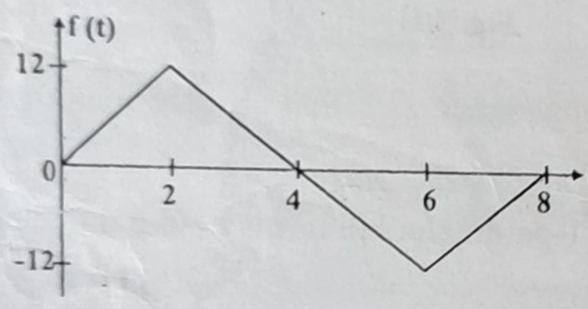
Duration: 3 hrs

Instructions: 1. Mention the question number clearly and write all the parts of a question at one place.
2. Draw circuit/figure wherever applicable.
3. Make suitable assumptions wherever necessary, symbols and notations have their usual meanings.

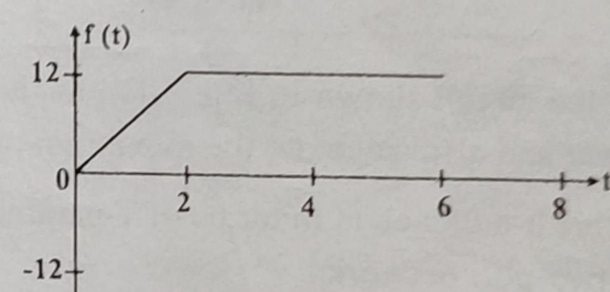
(Answer any Four)

1. (A) Write the expressions for the following signals using basic signals. Show the construction steps graphically for each signal. [8]

(i)



(ii)



(B) The circuit in Fig. 1(B) is initially under steady-state condition with switch S at position 1. The switch is moved from position 1 to position 2 at $t = 0$. Find the expression for the current i(t) after switching. [6]

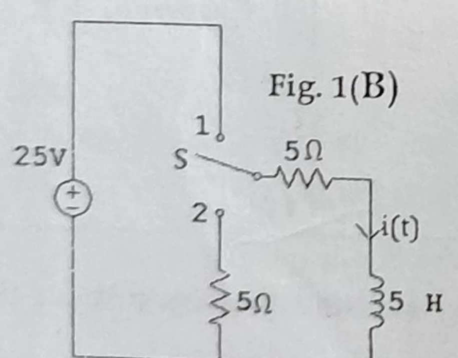


Fig. 1(B)

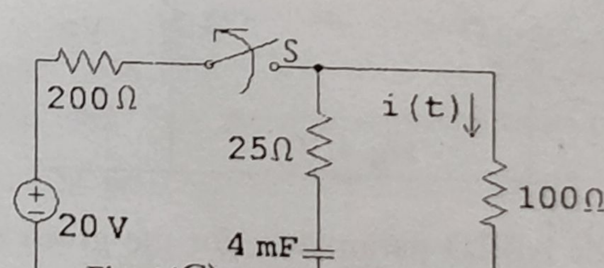


Fig. 1(C)

(C) In the circuit shown in Fig. 1(C), determine the expression for the current 'i(t)' for '$t > 0$'. The switch 'S' has been closed for a long time and is opened at '$t = 0$'. [6]

2. (A) Determine the node voltages V1 and V2 in the given circuit, Fig. 2(A). [8]

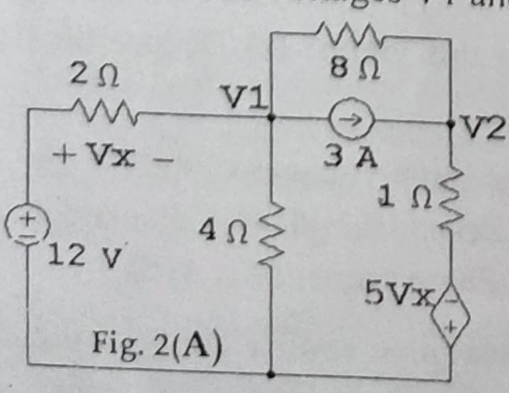


Fig. 2(A)

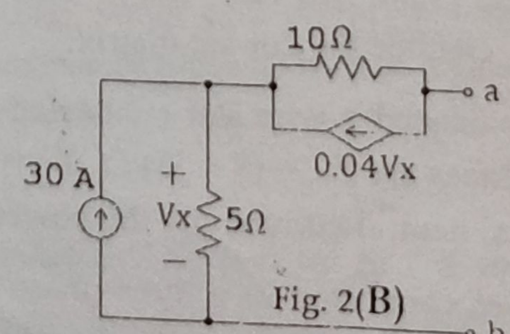


Fig. 2(B)

(B) For the circuit shown in Fig. 2(B), obtain the Thevenin's equivalent at the terminals a-b. [6]

(C) Given $F(s) = \frac{5(s+1)}{(s+2)(s+3)}$ [6]

(a) Use initial and final value theorems to find $f(0)$ and $f(\infty)$, where $f(t) = \mathcal{L}^{-1}[F(s)]$.

(b) Verify your answer in part (a) by finding $f(t)$ using partial fraction expansion.

3. (A) In the network shown in Fig. 3(A), determine the variation of current in 'R' when it is varied from 2 Ω to 10 Ω. [10]

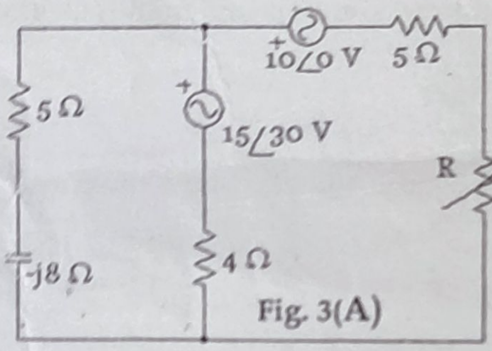


Fig. 3(A)

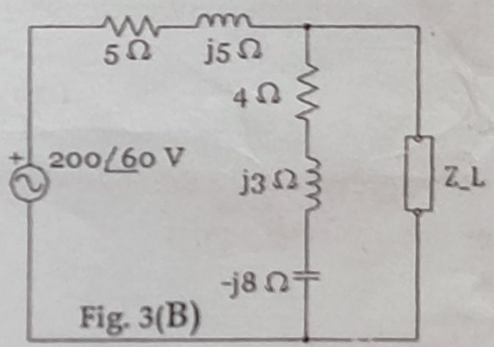


Fig. 3(B)

(B) For the circuit shown in Fig. 3(B), evaluate load impedance Z_L that absorbs maximum power and also calculate the maximum power. [10]

4. (A) Derive h-parameters in terms of T-parameters, and T-parameters in terms of Z-parameters for two-port networks. [6]

(B) Determine the Z and Y parameters of the network shown in Fig. 4(B). [8]

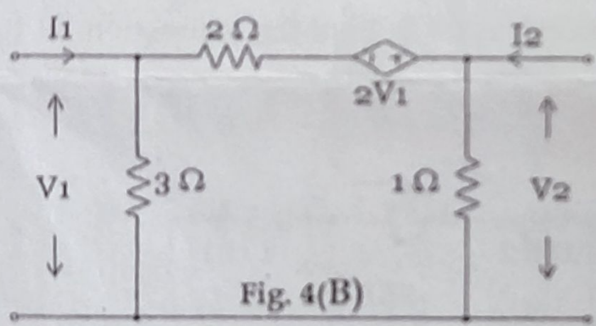


Fig. 4(B)

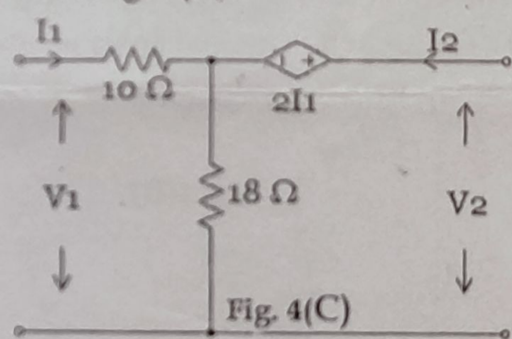


Fig. 4(C)

(C) Find the ABCD parameters for the given network in Fig. 4(C) and thereby test for the reciprocity of the network. [6]

5. (A) The reduced incidence matrix of an oriented graph is given: [8]

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ -1 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 & 1 \end{bmatrix}$$

(a) Draw the graph. (b) How many trees are possible for this graph? (c) Deduce the tie-set matrix. (d) Deduce the cut-set matrix.

(B) An unbalanced 4-wire star connected load has balanced line voltages of 420 V. The load impedances are $Z_A = (8 + j3)\ \Omega$, $Z_B = (4 - j5)\ \Omega$ and $Z_C = (10 + j6)\ \Omega$. Calculate the line currents, neutral current and the power in each phase. Phase sequence is ACB. [8]

(C) The load connected to a three-phase supply comprises three similar coils connected in delta. The line currents are 21 A and the kVA and kW inputs are 30 and 18, respectively. Find the phase voltages, the kVAR input and the resistance & reactance of each coil. [4]